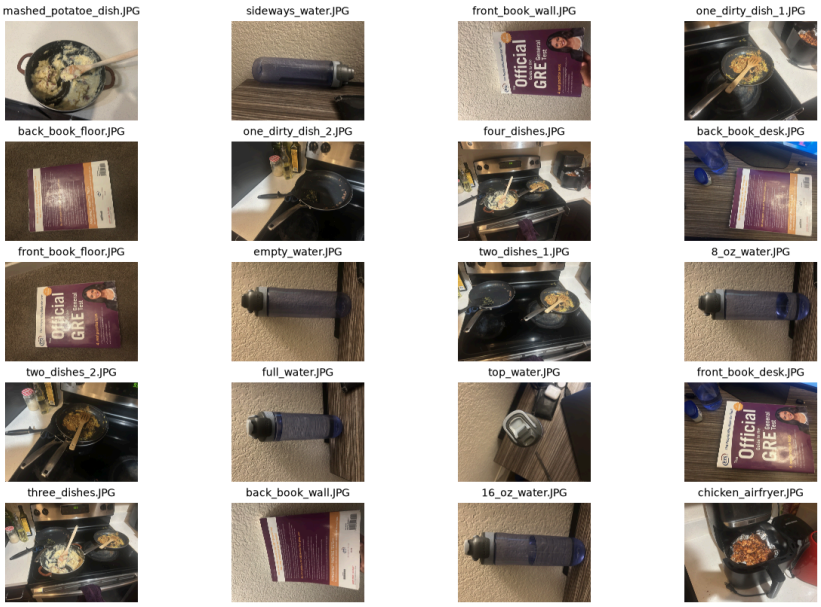
Assignment item	Points	Location in report	Extra credit?
Report Quality	5	-	-
Image Collection and Inspection	15	Section 2, p. 2	-
CLIP Image Embeddings	20	Section 3, p. 2	-
Text-to-Image Retrieval	20	Section 4, p. 2-3	-
Success and Failure Analysis	20	Section 5-6, p. 3–4	-
Image to Image Retrieval	5	Section 7, p. 4-5	-
Nearest-Pair Comparison	5	Section 8, p. 5-6	Yes
Retrieval Margin	5	Section 9, p. 6-7	Yes
Clustering Visualization	15	Section 10, p. 7-8	-
Total attempted before extra-credit conversion	**110**		
Maximum possible score after conversion	**105**		

CLIP Semantic Image Search

In this project, I evaluate the CLIP image encoder across a variety of challenging scenarios and build a semantic search. I created a dataset of 20 images with the main groups: dishes, water bottles, and books. In this paper I will evaluate CLIP's text to image retrieval, image to image retrieval, nearest pair comparison, retrieval margins, and group clustering on this dataset.

The purpose of the water bottle group was to test CLIP's ability to differentiate between different orientations of the bottle and how much water was in the bottle. I took pictures of the bottle when it was sideways, upright and at a birdseye view, mixed with the bottle being completely full, to partly full, to empty. The goal of the book group was to see CLIP's ability to differentiate between different backgrounds and different sides of the books. I took a combination of pictures of the front and back of the book with backgrounds of my carpet, desk, and wall. Lastly, the dishes group was intended to test clips ability to count the dishes as well as determine the contents of the dish. To take all of these pictures I used my IPhone 14.



I used the model the ViT-B/32 for my inferences. To preprocess my images I used the given preprocess function which resizes, crops, and normalizes the image. The shape of a single preprocessed image tensor is (1, 512) and the shape of the entire embedding matrix is (20, 512) as it is just 20 image tensors. Before normalizing the embeddings some of the norms were 10.455, 11.27, and 9.21, but after normalizing the norms were all 1.0. The time required to embed the 20 images was 288601000ns or 0.288601s, this is just encoding the image and normalizing the embeddings, not preprocessing.

The search function had two input parameters, a required query string and an optional integer k which told us to display the top k results by similarity score. First I tokenized, encoded, and normalized the text to get the embedded vector using the clip.tokenize and the encode_text functions provided by

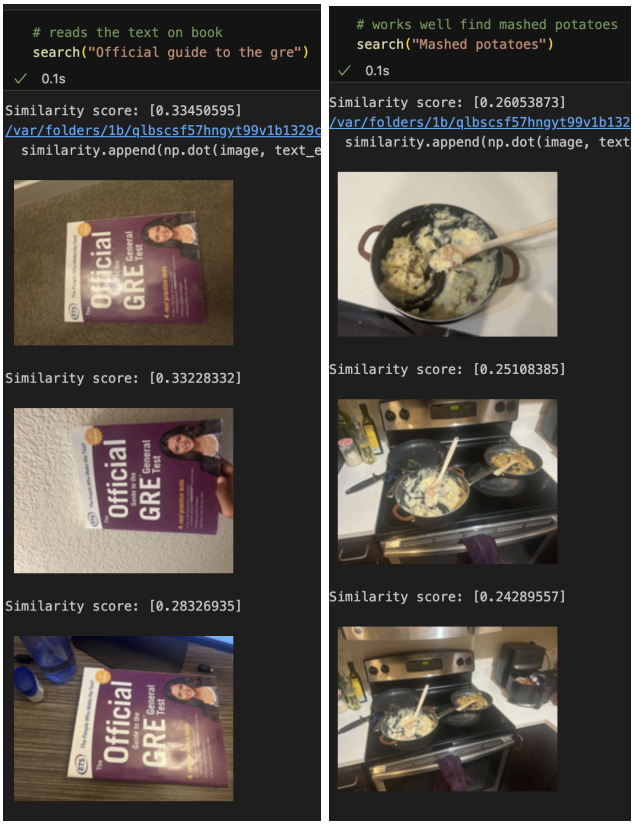
CLIP and used the pytorch norm function to normalize . Now all I had to do was find the most similar image embeddings to the query text embedding. I looped through the image embeddings and got the dot product of the image embedding and the text embedding. Then I sorted the list of the dot products and the image paths simultaneously so that similarity would go in descending order and image paths would be at the correct index to correspond to the similarity scores. Then I just looped through the top k scores and printed the similarity score and displayed the image using the Pillow Image.open, Image.thumbnail, and IPython.display functions.

The main failures that I was able to identify all centered around the water bottle group. When creating my dataset I wanted to test and see if CLIP would be able to tell the difference between a full, ⅔ full, ⅓ full, or empty bottle as well as the orientation of the bottle. To my surprise though clip was majorly confused on both factors and was not able to find the empty water bottle when queried for “empty water bottle”. My hypothesis is that since the water bottle is transparent there is only a slight edge to show how much water there is, and is otherwise hard to tell at any other point in the water bottle. I think CLIP is not able to find this edge leading it to mainly focus on the water bottle aspect of the query leading it to return an image of the bottle with very close similarity scores. In the dataset there were two images of completely empty water bottles shown on the right (top and bottom image) which are ranked 3rd and 5th respectively, while on the left are the two images that the query found most similar.

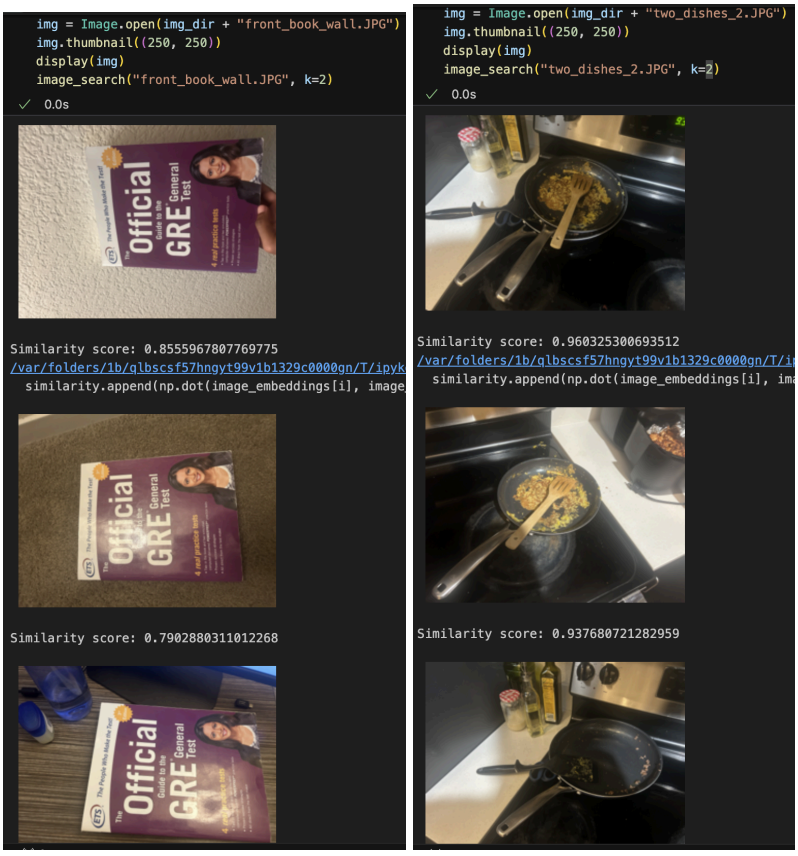


The other two groups that I had, books and dishes, seemed to work well as CLIP was able to identify many different aspects like background, count, words, and the items themselves. A good query that I found for my books group was “Official guide to the gre” which is the title of the book. CLIP found

the 3 images I had of the front cover of the book as the most similar. The other images of the book that I had were of the back cover that did not contain the title of the book which shows that CLIP was able to read it. For my dishes group the query “Mashed potatoes” which was the food in my pot in the pictures. I had multiple pictures of this pot with various other things in the image and CLIP was still able to identify the mashed potatoes among the background noise.



For image to image retrieval it appears that the nearest neighbors are driven by object identity. In my first query I tried to use an image of the front of my book against the wall to see if CLIP would identify the background wall as an important feature causing an image of the back of the book against the wall to be most similar. This did not work though as CLIP likely read the title of the book causing it to find the image with the front of the book to be most similar. Something that may change this is if I took the picture from further away so that the background wall took up more space and the book appeared tinier. In my second query I tried to see if CLIP would be able to identify there being two dishes as a prominent feature instead of the dish that was front and center. This was not the case though as CLIP found the pan that was front and center to be most important, and thus the most similar images were ones that featured the same pan. Something that may have changed the outcome of this test was if the image of two pans had the pans side by side instead of stacked, but I still believe that CLIP would have just identified the larger pan as the more important feature.

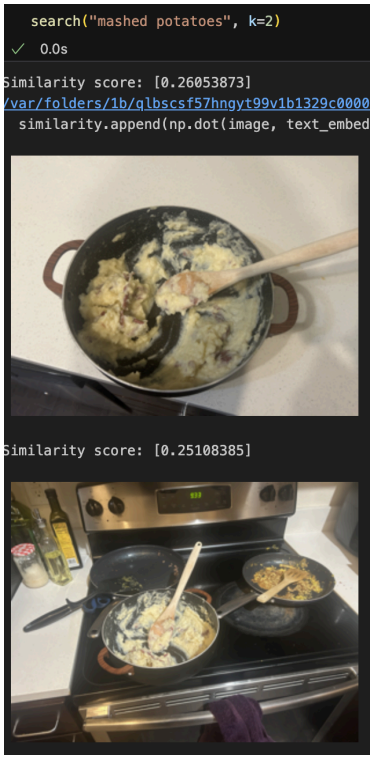


The top 5 most similar pairs revealed an interesting result when compared to my last tests. So far in my test CLIP has identified the most prominent object as the biggest tell for similarity, but in this list the background is used more. In the first four images shown below it holds true that the most prominent object is the same in both images, as they have the same dishes and water bottles in both. When you take a look at the fifth image though you see that the background is weighted more highly. These are images of different dishes, but the dishes are in the same location meaning that the background bottles, counter, and stovetop are exactly the same, which CLIP identifies making the similarity score go up.

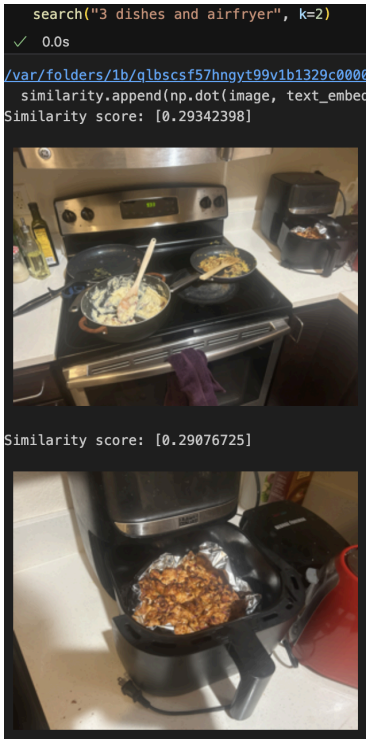


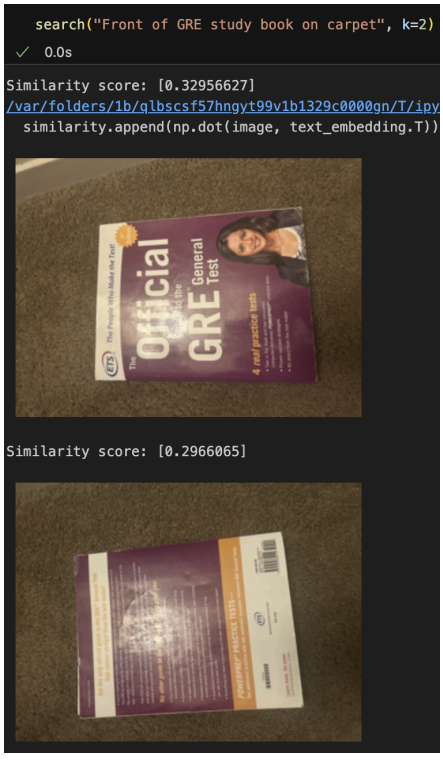
In the images below displaying the retrieval margins we can see the ambiguous rankings on the top row and the more certain rankings on the bottom row. On the top row, which all have a less than 0.01 difference in similarity score, we see that there are multiple images that contain the keywords of the text query in the image. For the first image the query is “mashed potatoes” and CLIP has clearly found the multiple instances of the pot filled with mashed potatoes. The same can be said about the second image with the query “empty water bottle” as CLIP identified “water bottle” to be the most important part of the query and found the multiple images of water bottles. The third ambiguous query, “3 dishes and airfryer”, is more interesting as CLIP has identified two images with similar similarity scores, but they focus on different parts of the query. The most similar image focuses on the first part of the query text, “3 dishes”, while the second most similar image focuses on the second part of the query text, “airfryer”. On the other hand, the bottom row of clear rankings don’t have an image that also focuses on the most important part of the text query. The first query, “front of gre book on carpet” is very specific, as there are only 1 image of the GRE book on carpet, but there are other options of the front and back on different backgrounds as well as the back on carpet. This led to a similarity score difference of approximately 0.03, meaning that CLIP was able to capture the entire text query which included two important features and identify the image that contained both features. Similarly with the text query, “Airfryer”, CLIP was very confident

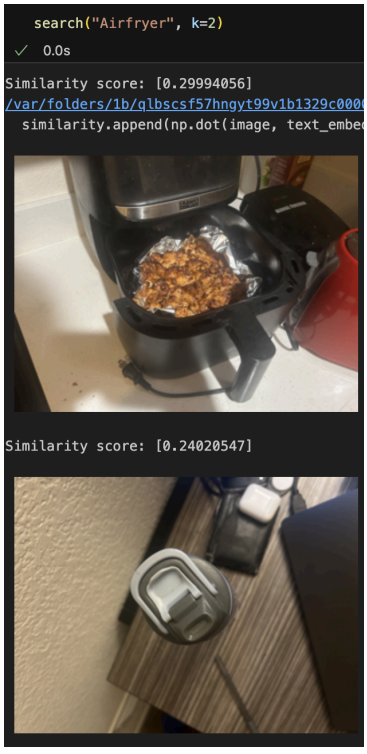
with its guess as there was an approximately 0.06 difference in similarity scores. This appears to be for a similar reason, there is only one other image with the airfryer, but in that image the airfryer is in the background, so CLIP chose the image with the airfryer front and center.











The clusters perfectly match the thematic groups I intended when designing my groups. The euclidean distances for each group are not as I expected, I would think that the book and water bottle group would have smaller euclidean distances since the images are of the same main object. While the dishes group would have a large euclidean distance as there are multiple pots, pans and an airfryer in the image. The euclidean distances further show the importance of the background, as the smaller euclidean distances of the water bottle group with 0.29 and the dishes group with 0.33, while the book group is at 0.47 are because of the backgrounds. The water bottle images are all taken with the same background being the edge of my desk and the wall and the dishes are all taken on the same countertop and stovetop, while the books are taken with 3 completely different backgrounds: a desk, carpet, and a wall.

